

Raffles Institution 2018 Preliminary Examination

H2 Biology

Paper 1

1.	D	7.	D	13.	D	19.	C	25.	C
2.	B	8.	C	14.	A	20.	C	26.	A
3.	A	9.	D	15.	C	21.	B	27.	A
4.	D	10.	A	16.	B	22.	B	28.	B
5.	C	11.	D	17.	D	23.	B	29.	A
6.	C	12.	D	18.	A	24.	C	30.	C

Paper 2

1 Fig. 1.1 shows the effect of low ATP on Phosphofructokinase (PFK) activity.

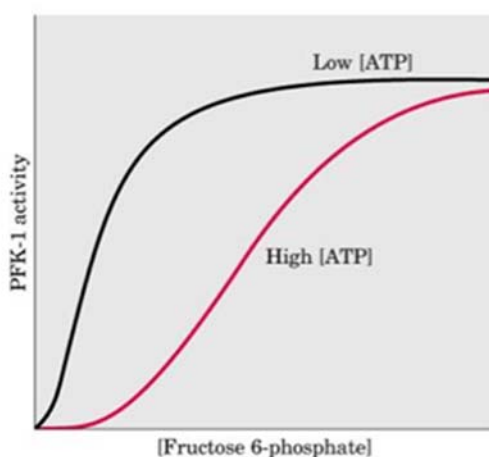


Fig. 1.1

- (a) (i) On Fig. 1.1, draw a graph to show the effect of high ATP on PFK activity [1]
- (ii) Explain the graph you have drawn. [3]
- 1) High levels of ATP inhibits PFK activity by lowering its affinity for fructose 6-phosphate.
 - 2) ATP* acts as an allosteric inhibitor which binds to an allosteric site which inhibits the PFK activity
 - 3) $[V_{max}]$: inhibition can be overcome by high substrate concentrations;
 - 4) [idea of cooperativity] binding of substrate to 1 subunit changes conformation of other subunits such that it becomes easier for substrate to bind;
- (iii) Name 1 molecule that will act as allosteric activator of PFK activity.[1]

AMP/ADP

R: fructose-6-phosphate

(b) Fig. 1.2 shows a PFK molecule.

(i) Describe the structure of PFK. [4]

- 1) PFK has a quaternary* structure made up of 4 polypeptide chains.
- 2) Each of which has primary structure which is unique number and sequence of amino acid in a polypeptide linked by peptide bonds.
- 3) Secondary, tertiary and quaternary structures are hence direct consequences of

- primary structure.
- 4) Secondary structure refers to regular coiling and folding/pleating of polypeptide held by hydrogen bonds between CO and NH groups of polypeptide backbone.
 - 5) Tertiary structure refers to a single polypeptide chain which is further folded into specific 3D conformation, held by bonds between R-groups within same polypeptide.
 - 6) Tertiary structure is maintained by hydrophobic interaction, hydrogen bonds, ionic bonds.
 - 7) Overall conformation of PFK is maintained by its primary, secondary and tertiary structure in addition to interaction among the 4 polypeptide chains.
 - 8) Presence of active sites* complementary in shape and charge to substrate.
 - 9) Presence of allosteric sites* complementary in shape and charge to activator/inhibitor.
 - 10) PFK is globular/water-soluble, arranged so that most of its hydrophilic amino acid side chains are on external surface of the protein while hydrophobic amino acid side chains are buried in interior.

(ii) Explain how a change in pH affect the PFK activity.[3]

- 1) Affects ionisation of R groups* of aa;
- 2) A change in pH will disrupt the ionic and hydrogen bond* that maintain the 3D conformation of active site of PFK;
(A: mention of either one of the bonds)
- 3) which will result in a change in 3D conformation of active sites* (R: catalytic sites), no longer complementary in shape and charge to substrate / substrate cannot fit, hence;
- 4) less enzyme-substrate complex* formed hence reducing the PFK activity;

[Total : 12]

2

(a) With reference to **Fig. 2.1** and **Table 2.1**,

- (i) identify the type of molecule as well as the type of regulatory sequences studied in the experiment.[3]

	Type of molecule	Position	Type of Regulatory Sequence
A	<u>Activator</u>	-2580 to -2400	<u>Enhancer</u>
B	<u>Repressor</u>	-1350 to -1000	<u>Silencer</u>
C	<u>RNA polymerase / General transcription factor</u>	-30 to 0	<u>Promoter</u>

- (ii) explain how you arrived at the identity of molecule B as well as the type of regulatory sequence at -1350 to -1000. [2]

1. <correct set of data comparison + quote data>
Wild type vs deletion construct 2 / deletion of -1350 to -1000, resulted in an increase of 282 a.u. in relative mRNA quantity. (A: deletion construct 2 resulted in increase in relative mRNA quantity of 550 a.u. compared to the wild-type construct of 268 a.u.).
2. Repressor* protein is unable to bind to silencer* sequence to lower transcriptional activity. (A: presence of -1350 to -1000 sequence allowed repressor* to bind to a silencer* sequence to downregulate transcription).
3. Thus, transcriptional initiation complex / RNA polymerase can assemble (more easily/readily) at promoter*, thus increasing frequency of transcription, which leads to more mRNA formed (marker's note : increase rate of transcription).

Note to marker: pt 1 is compulsory

- (b) The *LCT* mature mRNA excluding its poly-A tail is much shorter than the *LCT* gene. Account for this difference. [2]

1. Due to non-coding sequences in gene such as introns*, which are removed during splicing* by the spliceosome.

2. The gene includes promoter* sequences which are not transcribed by the RNA polymerase.

R: enhancer / silencer (not part of gene)

- (c) With reference to the *lac* operon, describe one difference in the regulation of lactose metabolism in bacteria and humans.[1]

	Eukaryotes	Bacteria
Effect of promoter	Each <u>gene</u> * is controlled by a <u>single promoter</u> *;	Multiple related <u>genes</u> * are controlled by the <u>same/single promoter</u> * (operon system);
Regulatory sequences involved	Regulatory sequences e.g. <u>enhancer</u> * and <u>silencer</u> * that involved in controlling frequency of transcription initiation may be <u>distal</u> ;	Regulatory sequences, e.g. the <u>operator</u> * that involved in control of transcription is <u>proximal</u> to the genes under its control;

- (d) (i) Complete the graph by showing the concentration of bacteria in culture over time from W to Z. [1]

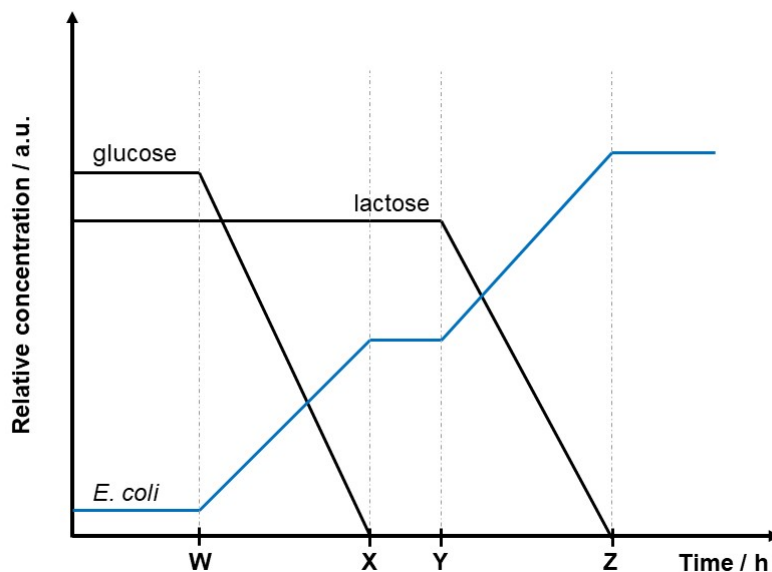


Fig 2.2

WX-Increase, plateau, increase

A: exponential increase

Using your knowledge of the *lac* operon, explain the shape of the graph for lactose from

- (ii) Time W to X [2]

1. Glucose present thus, low cAMP* levels;
2. Catabolite Activator Protein* (CAP) not activated, hence no CAP binding to CAP binding site in promoter and thus no up regulation of transcription of lac operon;
3. Beta-galactosidase* not expressed to break down lactose, hence lactose amount maintained;

OR

Permease* not expressed for lactose uptake, thus lactose level remains

unchanged;

(ii) Time Y to Z [4]

1. Lactose is converted to allolactose* which acts as an inducer*, binds to repressor*, repressor is inactivated;
2. As binding causes a change in conformation such that repressor no longer binds to the operator*;
3. RNA polymerase can bind promoter* and initiate transcription* of the structural genes (Lac Z, Y and A) to take place.
4. In absence of glucose, high cAMP* levels
5. result in cAMP-Catabolite Activator Protein* (CAP) complex which attaches to CAP-binding site on promoter strengthening affinity of RNA polymerase for promoter → upregulation of transcription of the *lac* operon
6. Beta-galactosidase* expressed in high amounts to break down lactose, hence lactose amount drops;

OR

Permease* expressed in high amounts for lactose uptake, thus lactose level remains drops;

[Total : 15]

3

(a) Discuss the role of centromere in the production of gametes. [3]

1. They allow (proteins called kinetochore proteins, and subsequently) spindle fibres, to attach; (MUST HAVE)
2. resulting in proper alignment of bivalents on equator in metaphase I/meiosis I;
3. so that homologous chromosomes can be equally separated to opposite poles during anaphase I/meiosis I;
4. resulting in proper alignment of chromosome singly in on equator in metaphase II/meiosis II;
5. so that sister chromatids / daughter chromosome can be equally separated to opposite poles during anaphase II/meiosis II;

Note : point 1 is compulsory and from points 2-5 any two. Without point 1, maximum marks that can be obtained is 2.

(b) The graph in Fig 3.1 shows the length of the spindle fibres during meiosis.

(i) In which regions of the graph did the centromeres detach from the spindle fibres. [1]
D and H

(ii) Which region on the graph corresponds to the stage shown in the Fig 3.2. [1]

F

(iii) Name and identify the 2 corresponding stages on the graph in Fig 3.1 that contributes to genetic variation in gametes, and explain how they bring about genetic variation. [6]

Stage :

1. Prophase I, A
2. Crossing over* of segments of non-sister chromatids of homologous chromosomes* at prophase I of meiosis I.
3. This leads to new combinations of alleles on chromosomes of gametes.

Stage :

4. Metaphase I, B

5. Independent assortment* of homologous chromosomes* occurs where arrangement of one pair of homologues at metaphase plate is independent of the arrangement of the other pairs of homologues;
6. This results in different combinations of maternal and paternal chromosomes in daughter cells;
7. there are 2^n possible combinations of gametes where n is number of homologous pairs. / 2^{23} possible combinations of gametes in human ;

A: Metaphase II, F;

only if answer clearly makes ref to;

random arrangement of /independent orientation of non-identical sister chromatids* (R:chromosome) of each chromosome at the metaphase plate & their subsequent separation during metaphase II and anaphase II respectively

- (c) (i) What is the mode of inheritance of the disease WAS? [1]

Sex-linked recessive ;

- (ii) Use suitable symbols to represent the alleles of the gene controlling the disease WAS [1]

X^W : Dominant allele that codes for no WAS on X chromosome

X^w : recessive allele that codes for disease WAS on X chromosome;

Must show both alleles

Must have X chromosome on symbol

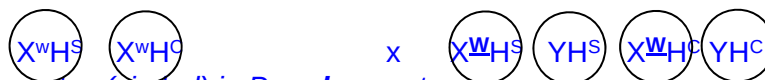
- (iii) Use a genetic diagram to explain the results of **couple 2**. [4]

1. [1] parental phenotype and genotype;

P phenotype: wavy-hair female WAS sufferer x wavy-hair normal male ;

P genotype: $X^wX^wH^S H^S$ x $X^WYH^S H^C$

Gametes (n):



2. [1] all correct gametes (circled) in P and punnet square

		Male gamete			
		X^wH^S	YH^S	X^WH^C	YH^C
Female gamete	X^dH^S	$X^wX^wH^S H^S$ Normal Straight hair female	$X^wYH^S H^S$ WAS sufferer, straight hair male	$X^wX^dH^S H^C$ Normal wavy hair female	$X^wYH^S H^C$ WAS sufferer, wavy hair male
	X^dH^C	$X^wX^wH^S H^C$ Normal wavy hair female	$X^wYH^S H^C$ WAS sufferer, wavy hair male	$X^wX^dH^C H^C$ Normal curly hair female	$X^wYH^C H^C$ WAS sufferer, curly hair male

3. [1] Punnett square with correct genotypes and related phenotypes

offspring genotypic ratio	$X^wX^dH^S H^S$	$X^wX^wH^S H^C$	$X^wX^wH^C H^C$	$X^wYH^S H^S$	$X^wYH^S H^C$	$X^wYH^C H^C$
offspring phenotypic	Normal Straight	Normal wavy	Normal curly hair	WAS sufferer,	WAS sufferer,	WAS sufferer,

ratio	hair female	hair female	female	straight hair male	wavy hair male	curly hair male
	1	2	1	1	2	1

4. [1] Correct offspring genotypic and phenotypic ratio

[Total : 17]

4

- (a) (i) With reference to **Fig. 4.1**, state the best conditions for the growth of cucumber.[1]
0.13% carbon dioxide, 30°C, above 5 arbitrary units of light;
- (ii) Define the term limiting factor.[1]
1. when a chemical process is affected by more than one factor, its rate is limited by that factor which is nearest its minimum value;
 2. It is that limiting factor which directly affects a process if its magnitude is changed;
AVP
- (iii) With reference to **Fig. 4.1**, explain reasons for the difference between curves B and C. [3]
1. Curve B reaches a plateau at rate of photosynthesis of 95 mm³cm⁻²h⁻¹ at lower light intensity of 3 units, whereas Curve C reaches a higher maximal rate of photosynthesis of 210 mm³cm⁻²h⁻¹ at higher light intensity of 5 units; (idea of : max rate being different, with values)
 2. due to presence of higher CO₂ concentration at 0.13 (graph C);
OR
 3. Curve B: reaches max rate at light intensity of 3 units, while Curve C reaches max rate at light intensity of 5 unit
 4. light intensity is no longer a limiting factor as any further increase in light intensity does not affect rate of photosynthesis. Hence, CO₂ concentration becomes limiting;
(max 2 from 1 and 2 or 3 and 4)
 5. Plant C achieve higher rates of photosynthesis as compared to B, because higher CO₂ / substrate concentration during carbon fixation* in Calvin cycle.
 6. results in higher frequency of effective collision* with enzyme Rubisco (and RuBP), thus more enzyme-substrate complexes* being formed;
- (b) With reference to **Fig. 4.2**,
- (i) explain why most plants are characteristically green, and [2]
1. Percentage of green light absorbed by (all 3 pigments – marker discretion) is very low at less than 5%;
 2. Green light is reflected / transmitted;
For reference: Blue ~ 490–450 nm, red ~ 635-700 nm
- (ii) explain the shape of the action spectrum. [3]

1. Peak(s) of action spectrum/ highest rate of photosynthesis, correspond to absorption peak(s) (Idea of :action spectrum = sum of absorption of all pigments);
2. Light absorbed by different photosynthetic pigments is used for photoactivation/ photosynthesis
3. thus higher percentage of light absorption leads to higher rate of photosynthesis in red and blue light; (idea of : chl a peak corresponding to action spectrum peak)
4. carotenoids / chl b absorbed at blue and red light, broadening the range of wavelength that can be used for photosynthesis
5. Region between 500 to 600 nm (R: green) is not an exact match between absorption and action spectra due to role of other accessory pigments;

(c) In conducting experiments to obtain the action spectrum for photosynthesis, explain why use of organic solvents such as acetone or ether is necessary to extract photosynthetic pigments. [2]

1. Chlorophyll pigments are membrane-bound (A: thylakoid and chloroplast membranes);
2. Only organic solvents can dissolve the membrane, that is composed of a phospholipid bilayer;

[Total: 11]

5

(a) Name the method of gene transfer seen in Fig. 5.1 and describe it briefly. [4]

1. Conjugation*;
2. Sex pilus* of F^+ bacterial cell makes contact with a F^- cell ;
3. and retracts / contract to bring the F^- cell closer so a mating bridge* is formed between the 2 cells; (R: shorten)
4. One of the 2 strands of plasmid DNA in F^+ cell is nicked and transferred from F^+ cell to the F^- cell through mating bridge (via rolling circle mechanism);
5. as the other DNA strand is used as a template for elongation of nicked strand;
6. The single strand F plasmid DNA circularizes in F^- cell and is used as a template* to synthesise a complementary strand;
7. for a double-stranded F plasmid DNA resulting in F^+ cell;

(b) After several hours of incubation, the cells were transferred to minimal medium (without methionine and threonine). Predict and explain if any bacterial growth would be observed.[2]

1. No growth;
2. Filter prevent physical contact between the 2 strains and did not allow for conjugation / did not allow for sex pili from F^+ cell to come into contact with F^- cell

(c) (i) Predict if the bacterial progeny from this tube can survive in minimal medium without additional of methionine and threonine.[2]

1. Progeny can survive in minimum medium;
2. DNA enclosed inside cell / mating bridge / not exposed to surrounded, and hence not degraded by nuclease;
3. Conjugation occurred, allowing genes coding for enzymes involved in metabolism of met and threo to be tranfered between the 2 bacteria.

R: reference to nuclease needed to nick the F plasmid for conjugation.

(ii) State the purpose of setting up this tube.[1]

To exclude transformation* as a possible method of gene transfer

(d) (i) With reference to Fig. 5.3, determine if strain A or Strain B is the F⁺ cell.[1]
Strain B

(ii) In cases of gene transfer shown in Fig. 5.3, the F plasmid-bacterial chromosome is usually not transferred completely to the recipient cell. Random breakage interrupts the process. The recipient cell is unable to go on to donate any genes to other cells. With reference to Fig. 5.3, suggest if the recipient cell is considered an F⁺ or and F⁻ cell after the transfer. Explain your answer.[2]

1. F⁻;
2. F factor is not transferred, so the cell is unable to synthesise sex pili to initiate conjugation; (A: incomplete F plasmid transferred)

[Total: 12]

6

(a) Describe two changes to the bones in the feet of horses that have taken place over the past 50 million years. [2]

1. Bones became larger/longer/thicker;
 2. Number of bones became fewer/idea of fewer toes or bones fused;
 3. Fewer bones touched ground/lower surface area in contact with the ground;
- Accept any other valid observable changes

(b) (i) Explain **one** advantage to *Eohippus* of the arrangement of bones in its feet. [1]

With a larger surface area (how arrangement of bones confer advantage) in contact with the ground, it can distribute its weight and not sink into the soft mud/ can escape predators (advantage);

(ii) Based on the information given, explain how the changes in the arrangement of the foot bones of horses support Darwin's theory of evolution by natural selection. [4]

Descent from a common ancestor

1. It shows descent with modification from a common ancestor, as basic underlying homologous foot structure was modified over time;

With modification

2. There was variation in size/number/arrangement of foot bones in population; (context specific)
3. Environment changed from swampy to hard and firm ground; (envt change, quote)
4. They faced selection pressure of predation*; (idea of selection pressure, quoting predation) R: if selection pressure is vague like "environment" or wrong like "hard/soft ground"
5. Natural selection selected for individuals who had larger/fewer foot bones as they are; (selected traits)
6. Able to run faster on firm ground to avoid predation; (suggest why trait is an advantage)
Pt 5 must be correct first.
7. Those who survive*, are able to reproduce and pass on their alleles for larger/fewer bones to the next generation;

- (c) (i) Fossils of *Eohippus*, *Mesohippus* and *Merychippus* have been unearthed only in the North American continent. Explain how the biogeographical distribution of these fossils support Darwin's theory of evolution. [3]

Accounting for their geographical distribution and common ancestry:

1. Their close geographical proximity to each other suggest that the 3 extinct horses descended from a common ancestor* that originated in North America/same continent*; (idea of center of origin of common ancestor)
2. Ancestral population dispersed and spread outwards from center of origin and colonized various habitats but confined by the oceans surrounding North American continent; accept idea of discontinuous distribution; (Distribution: all extinct horse species are only found in N America and nowhere else)

Accounting for descent with modification

3. Occupying different niches/environments, with natural selection, they evolve over time to give rise to different species; (Speciation from common ancestor)
4. Their fossils/structural foot bone changes demonstrate descent with modification as it changes over time; (descent with modification);

- (ii) Explain how scientists created the phylogram in Fig. 6.2? [2]

1. They studied DNA sequences of a homologous gene e.g. cytochrome C found in all the organisms;
2. Through sequence alignment and comparing difference in nucleotides, they can work out relationship between organisms; R: genotype, analyse
3. The fewer nucleotide differences, the more closely related the species and therefore they share a more recent the common ancestor/branch point more recent in time;
4. Carbon dating of fossils will tell how long ago an extinct species existed so that you can place them along a timeline;
5. Changes in nucleotide sequence accumulate over time with clockwork regularity. We can thus estimate the timeline of speciation events;

[Total :12]

7

- (a) With reference to the steps in Fig 7.1, identify the steps and explain the processes that kills the pathogenic bacteria.[2]

1. Step 4 - Fusion of phagosome with lysosome* forming phagolysosome;
2. Step 5 - Lysosomal / hydrolytic / digestive enzymes digest pathogen into harmless substance/cuts up antigens into short peptides; Thus bacteria is unable to further infect the host;

- (b) Describe how the humoral immune response enhances the action of the macrophage on the pathogenic bacteria.[3]

1. Plasma cells produce antibodies/immunoglobulins;
 2. Agglutination*, where antibodies clumps bacteria to promote phagocytosis* by macrophages;
 3. Opsonisation*, where antibodies tag pathogen to promote phagocytosis* by macrophages;
 4. Fab region of antibody/immunoglobulins binds to antigens on pathogen;
 5. Fc region binds to Fc receptor on phagocytes to promote phagocytosis;
- (max 3 marks)

- (c) Explain how *Mycobacterium tuberculosis* protects itself from the host immune system.[2]

1. *M. tuberculosis* has cell walls that contain mycolic acids*; (KIV if MUST HAVE)
2. Protects bacterium from acids / digestive enzymes / lysosomal enzymes / reactive

oxygen species produced by lysosome;

3. inhibit the fusion of the phagosome with lysosomes, hence no phagolysosome is formed and no lysosomal enzymes are available to kill the bacteria;
(max 2 marks)

- (d) With reference to Fig 7.2, explain how rifampin acts to achieve bactericidal effects on *Mycobacterium tuberculosis*. [3]

1. Rifampin is a competitive inhibitor of RNA polymerase;
2. Structure of rifampin is complementary in shape/conformation to active site of RNA polymerase;
3. Blocks active site*, prevents binding of free ribonucleotides / DNA template/promoter.
4. prevents transcription* of bacterial genes hence no mRNA produced; (R; transcription of mRNA)
5. Prevent expression of essential/housekeeping genes / no proteins produced;

[Total:10]

8

- (a) With reference to **Fig. 8.1**, suggest why the sea ice extent in March is different from that in September. [1]

Ice extent was higher in March than Sept, as March is during winter/colder months where sea ice forms/less melting and Sept occurs during summer/warmer months where more ice have melted.

Temperature (hot/cold) / season difference (winter/summer) between March and September
→ More melting / more freezing → difference in sea ice extent

- (b) (i) Identify X shown in **Fig. 8.2**. [1]

X: duration of sea ice melt;

- (ii) Describe how X changes from 1979 to 2015. [1]

1. X increased by about a month from 1979 to 2015
2. Sea ice melt season is starting earlier and ending later , melt season spanning from mid june to mid sept in 1979, to late may to late sept in 2015;

- (iii) Explain how this is linked to the changes in sea ice extent as shown in Fig. 8.1 and Fig. 8.3. [3]

Fig 8.1

1. Longer sea ice melt season results in more ice melting, [E]
2. lowering minimum sea ice extent observed in September (/min); [O]
3. less time is available for formation of sea ice in winter months; [E]
4. decrease in maximum sea ice extent seen in March (/max) [O]

Fig. 8.3

5. Longer duration of sea ice melt resulted in less ice remaining frozen past melt season;
[E]
6. As seen in decrease in the amount / proportion of multi-year old ice / decrease in extent of sea ice over all ages; [O]
7. Which implies thinning of ice cover over the years;

Mark allocation:

Max 2 for each Fig

Max 2 for Obs across Fig

(c)

(i) State the class of receptors that melanopsin belongs to.[1]
G-protein coupled receptor;

(ii) With reference to Fig. 8.5, describe how phospholipase C could become activated.[3]

1. Light causes a conformational change in the cytoplasmic domain of melanopsin;
2. Allowing melanopsin to bind to an inactive G protein;
3. Causing exchange of its bound GDP for a GTP, activating G protein;
4. Activated G protein moves along cytoplasmic side of membrane;
5. To bind to and activate phospholipase C;

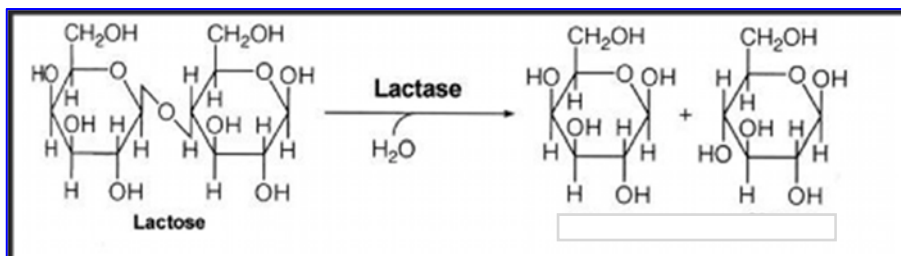
[Total : 15]

Paper 3

Section A

1

- (a) Show how a lactose molecule is hydrolysed into its monomers in the space provided below. [2]



1. Shown* and labelled lactose structure hydrolysed to form correct structure of monomers;
2. Shows how water is involved in hydrolysis;
[A: if carbon not drawn]

- (b) Compare the structures labelled **A** and **B** in Fig. 1.2. [3]

1. A= alpha helix and B=beta pleated sheet

Similarities

2. Both maintained by hydrogen bonds* between NH and CO* of polypeptide backbone*;
3. Both consist of amino acids* joined together with peptide bonds*;

Differences

4. A is helical in shape whereas B comprised of flat sheet which becomes folded (R: pleated sheet does not describe);
5. A is formed within a single region of polypeptide whereas B may be formed between different regions of same polypeptide
6. In A, hydrogen bonds are formed between O and CO group of one amino acid and H of NH group of another, 4 amino acids away, but in B, hydrogen bonds are formed between CO group of 1 region and NH gp of an adjacent region

[For full marks, must have 1 similarity + 1 difference]

- (c) (i) Explain a similarity and a difference in structure between the fungus lactase and the human lactase. [4]

Similarity [2]

1. Active sites* of both fungal and human lactase have similar 3D conformation and are complementary* in shape and charge to lactose;
2. So as to allow for hydrolysis of glycosidic bonds* in lactose;

Difference [2]

3. Consists of different structural amino acids in polypeptide of fungal and human lactase (A: primary structure in place of amino acids);
4. Both have different 3D conformation due to having different bonds, e.g. ionic / hydrogen bonds;
5. Hence they have different optimum pH because the bonds are stable at different pH;

- (ii) Explain why the fungal lactase will only work in the stomach but not in the small intestine. [3]

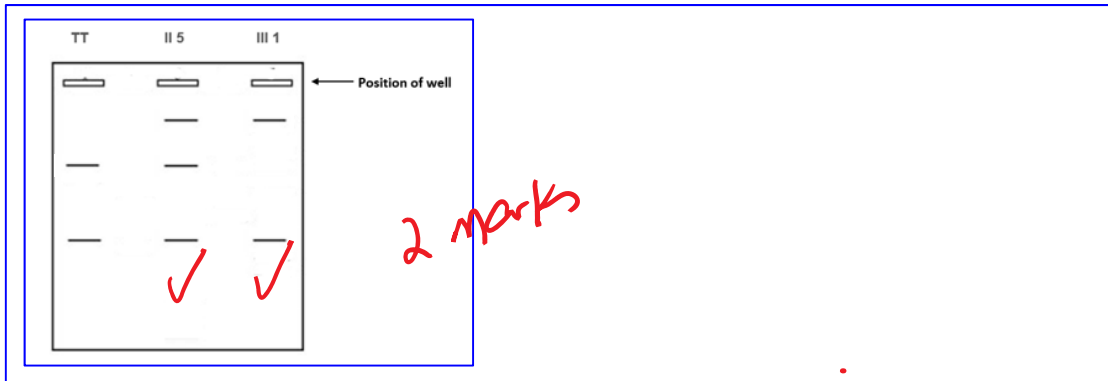
1. Stomach has an pH of acidic around 2, which is within pH range of fungal lactase;
Or
Small intestine has an alkaline environment, which is out of pH range of fungus lactase;

2. Deviation from optimum pH, results in lowering of rate of reaction as excess $[H^+]$ or $[OH^-]$ ions may affect ionisation of R-groups of amino acids;
3. Results in change in specific 3D conformation/shape of enzyme's active site or results in causing enzyme to be denatured*;

- (d) (i) With reference to Fig. 1.3, discuss the effectiveness of lactase treatment in reducing the symptoms of lactose intolerance. [3]
1. Lactase treatment was effective in decreasing amount of H_2 in breath test;
 2. Compare placebo and lactase-treatment, e.g. steep/sharp vs gradual/slight increase, highest vs lowest amount;
 3. Little / slight difference between high dosage and low dosage of lactase;
 4. + data quoted to support points 2 and 3, e.g. max value, difference in H_2 amount etc;
- (ii) Explain the shape of the graph for the group given placebo. [2]
1. From 0 to 3 hours: sharp / steep* increase in amount of H_2 produced from 0 to 56 ppm; (shape and paired values required)
 2. Due to lack of lactase resulting in large amount of lactose reaching large intestine, hence high rate of lactose fermentation;
 3. At 3 hours, max reached at 56 ppm / from 3 to 6h, decrease slightly / gradually* to 53 ppm; (shape and paired values required) R: plateau
 4. Since no more H_2 produced as all lactose was used up in fermentation;
- Must have 2 or 4 to get max 2
- (iii) Draw on the graph in Fig. 1.3, the results obtained from this participant's test. [1]
- 0 to 3 hours: start at higher +
3 to 6 hours: follows trend of lactase or plateau (R: gradual increase that follows placebo);
- (e) Describe a mechanism that accounts for this age-dependent lactase regulation during adulthood. [2]
1. DNA methylation / histone deacetylation / chromatin remodelling complex, condensation of DNA / chromatin;
 2. Prevents access of transcription factors and RNA polymerase / transcription initiation factors to promoter, decrease frequency of transcription;
- R: repressors binding to silencers since this is not long-term control of gene expression
- (f) (i) State the mode of inheritance of lactase persistence. [1]
- Autosomal dominant;
- (ii) Explain how you arrived at your answer in (f)(i). [4]
- Dominant
1. **Individual II5 and II6 are both LP. However, some of their children, II5 and III7 are non-LP; (must clearly state the individuals and their phenotype)
 2. Shows that II5 and II6 are heterozygous where non LP allele is masked by presence of LP allele;
 3. If it is recessive, then all their children will have LP;
- Autosomal
4. **Father with LP (II6) produce daughter who is non LP (III5); (must clearly state the individuals and their phenotype)
 5. III5 inherited X chromosome from II6 but is non LP. Therefore LP allele is not found on X chromosome;
- Or
- If it sex-linked and dominant, father with LP would produce daughters that are all LP;

- (g) Fig. 1.5 shows a gel with the results of the RFLP analysis for an individual who is homozygous for the T allele. Draw on Fig. 1.5 to show the results of the banding patterns of individuals II 5 and III 1 from Fig. 1.4. [2]

Answer:



- 2 Fig. 2.1 shows the principle of ELISA.

- (a) Explain how the structure of the patient's Ig enables its binding to the NS-1 that coats the wells of the polystyrene plates. [2]

1. Each Ig contains 2 antigen binding site;
2. Complementary in shape and charge to antigen/epitope of NS-1;

- (b) With reference to Fig. 2.1 and using your knowledge of enzymes, explain how the process of ELISA allows quantification of the concentration of human Ig against NS-1. [3]

1. As concentration of human Ig against NS-1 bound to NS-1 on the well increases;
2. concentration of secondary antibody tagged with enzyme increases;
3. Increase in enzyme concentration results in increase frequency of effective collision/enzyme substrate complex formation/results in increase in rate of reaction between chromogen and active site of enzyme;
4. Increase in intensity of blue product formed within fixed time (idea: rate), which is detected using a spectrophotometer;

- (c) (i) Describe one characteristic of primers used and how it facilitates qPCR. [2]

1. Single-stranded;
2. Complementary to 3' end viral RNA sequence so that they can anneal to viral RNA;
3. Contain a free 3'OH end so that
4. Tag polymerase can extend the DNA/synthesize complementary DNA strand/form phosphodiester bonds between adjacent deoxyribonucleotides;

- (ii) Suggest how SYBR® green interacts with the PCR products. [1]

1. Sybr green intercalates/bind itself in space between the base pairs;
Reject reference to formation of hydrogen bonds with nitrogenous bases/complementary base pairs.
2. Structure of Sybr green is complementary in shape to grooves of eg: DNA-RNA hybrid;

- (d) (i) Explain the purpose of including the control with no RNA. [1]

1. This is a negative control to show base line fluorescence level if there is no amplification of viral RNA/PCR product formed;

AVP; ORA;

- (ii) If a typical qPCR cycle takes 3 minutes, calculate the minimum time taken to report a positive result.

Minimum time taken is

[2]

1. 3 min x 10/11 cycles; (Ans must show working and number of cycles needed)
2. 30/33 minutes;

- (iii) Suggest why qPCR is a better test compared to ELISA. [2]
1. Allowing earlier detection of dengue infection;
 2. Viral RNA is immediately present in infected patient whereas it take some time before Ig is produced;
 3. Faster than ELISA;
 4. with reference to 30 min vs 3 hour;
 5. AVP
- (e) (i) describe and explain the temperature trends in Singapore from 1974 to 2017, [2]
1. Annual temperatures increased from 26.0°C in 1974 to 27.7°C in 2017 (A: +/- 0.1°C);
 2. Globally, industrialisation / burning of fossil fuels for electricity/ deforestation led to increased release of CO₂;
 3. Increased greenhouse gas emissions (GHG) led to increased trapping of radiation/heat causing temperatures to rise; (R: radiation reflected from/to earth)
- (ii) explain the impact of climate change on transmission of dengue in Singapore. [2]
1. Global warming results in higher temperatures, thus conditions become more suited for survival of mosquitoes / shortens the mosquito's life cycle ;
 2. Transmission of dengue virus (DENV) will increase if mosquito vector, *Aedes aegypti*, that transmits dengue is present / increases as the virus reproduces within the mosquito ;
 3. Faster replication of dengue virus in the mosquito; (ref: shorter incubation period in mosquito)
- (iii) With reference to Fig. 2.4a and 2.4b, and the information given above, comment on the aspect of climate which has a greater influence on dengue transmission in Singapore. [2]
1. Temperature has greater influence over dengue transmission;
 2. During June to October, dengue infections were high, which correlates with higher temperatures May to Oct, where as rainfall was low/moderate in those months.
- (f) (i) Given the above information, explain how nuclear division may be affected. [2]
1. Paternal chromosomes did not condense during prophase*;
 2. Mitosis is arrested, and thus unable to produce daughter cells;
 3. Chromosomes are unable to align singly at the metaphase plate during metaphase*;
 4. Non-disjunction occurs / sister chromatids of paternal chromosome are unable to separate during anaphase*;
 5. Paternal chromosomes may get entangled during anaphase*, and thus resulting in DNA damage;
- R: meiosis, and cannot form gametes.
- (ii) Given the above information, justify Singapore's choice to adopt the release of only *Wolbachia*-infected males. [2]
1. Male mosquitoes do not bite, whereas release of female mosquitoes will increase mosquito bites;
 2. It helps to bring down mosquito population, as the cross of infected male with wild type does not result in new mosquito offsprings;
 3. Conversely a cross with infected female mosquitoes will still result in propagation of mosquitoes albeit they are unable to transmit dengue;
 4. Idea of unable to control mosquito population if there is undesirable outcome eg: ecological damage;
 5. AVP;
- R: prevents the transmission of dengue

Section B

- 3 (a) Account for the importance of hydrogen bonds in biological organisms. [13]

Context	Importance
Protein structure	Ref to levels of protein structure 1. <u>Secondary structure</u>* maintained by hydrogen bonds between <u>CO*</u> and <u>NH*</u> groups of peptide backbone; 2. <u>Tertiary structure</u>* maintained by hydrogen bonds <u>between R groups*</u> of same polypeptide to allow for protein to fold into functional conformation; 3. <u>Quaternary structure</u>* maintained by hydrogen bonds between <u>R groups between polypeptides*</u> allowing for interaction between different subunits of same protein; Relate structure to function 4. Ref. to specific example: Collagen: hydrogen bonds between R groups of adjacent polypeptide in a tropocollagen 5. → high <u>tensile strength</u>; Note: tensile strength (pt5 and pt 10 – 1 mark) 6. A: General description 3D conformation eg. Enzyme active site 7. Ref. to → crucial for function eg binding to substrate complementary in shape and charge to catalyse reaction 8. Ref to co-operativity eg: allosteric enzyme, haemoglobin
Cellulose/Carbohydrates	9. Cellulose: <u>hydrogen bonds between cellulose chains to produce cellulose fibres</u>; 10. that has <u>high tensile strength</u>;
Solubility	11. In a largely aqueous environment, <u>ability of molecules to form hydrogen bonds with water</u>; 12. a. <u>enables solubility</u> so that molecules can <u>e.g. (give 1 e.g.):</u> transport, diffuse through, metabolised; b. lack of hydrogen bonds result in insolubility – relate to function eg. Structural.
DNA structure	13. Hydrogen bonds <u>between complementary bases</u>; 14. causing DNA to be a <u>stable double-stranded</u> molecule;
DNA replication / transcription	15. Hydrogen bonds between <u>complementary bases (A: idea of 2 strands)</u> are <u>weak bonds</u>; 16. hence it is <u>easy break and separate DNA into 2 strands</u> to allow for <u>DNA replication / transcription</u>; 17. DNA <u>replication</u> uses 1 strand as a template to synthesise new strand: <u>Free nucleotides / deoxyribonucleoside triphosphate (dNTPs)</u> form <u>hydrogen bonds</u> with bases on DNA strand; 18. <u>Transcription</u>: same as above but <u>free ribonucleotides / ribonucleoside triphosphate</u> form <u>hydrogen bonds</u> with DNA strand; For 17 and 18, should identify product as either DNA or RNA; 19. Function: to use 1 strand as <u>template*</u> to synthesise a <u>complementary strand</u>;
tRNA and Translation	20. tRNA has <u>intramolecular</u> hydrogen bonds; 21. maintain <u>3D conformation</u> of tRNA so that tRNA can <u>fit</u> into <u>active site*</u> of <u>aminoacyl-tRNA synthetase</u> to be joined to its corresponding amino acid;

	22. Translation: <u>anticodon</u>* on <u>tRNA</u>* forms hydrogen bonds with <u>complementary</u>* <u>codon</u>* on <u>mRNA</u>*; 23. Allows polypeptide synthesised to have <u>amino acid sequence corresponding to codon sequence</u> on mRNA;
Ligand-receptor interactions	24. <u>Ligand-receptor</u> interactions may also involve hydrogen bonds allow ligand to <u>bind</u>, cause conformational changes and trigger <u>cell signalling</u>; 25. As hydrogen bonds are <u>weak interactions</u>, it will allow for ligand to <u>bind and dissociate</u> (A: bind temporary) from receptor;
Enzyme-substrate complex	26. Interaction involving enzyme <u>active site</u>* and <u>substrate</u> may also involve hydrogen bonds 27. which would help to <u>bind to substrate</u> and hold them in <u>correct orientation</u> for reaction;
Transpiration	28. Hydrogen bonds form <u>between water molecules</u>; 29. allow for <u>water uptake</u> in plants via <u>transpirational pull/ capillary action</u>;

AVP;

QWC: ref to 4 different context;

(b) Discuss the diverse roles of membranes in a cell. [12]

1. Membranes are made up phospholipid bilayer with hydrophobic core;
2. Protein being embedded within bilayer;

Context	Roles
Selectively permeable / S	3. Being <u>selectively permeable</u>, membrane only allows some molecules to cross directly without assistance; 4. <u>Non-polar/uncharged</u> (ignore: hydrophobic) molecules are able to dissolve and <u>diffuse</u> through; 5. <u>Ions/charged</u> and/or most <u>polar</u> molecules (ignore: hydrophilic) are <u>repelled</u>; (Generic principle cannot be replaced by H⁺)
Compartmentalisation / C	6. Formation of <u>unique environment</u> for <u>highly specialised activities</u>; 7. e.g. <u>lysosomes</u> maintain an <u>acidic</u> environment that favours its enzymes to work; 8. <u>localization</u> of enzymes for <u>reactions</u> to take place so that they are not suppose to catalyse reactions that they are not supposed to; 9. e.g.: <u>enzymes</u> in Calvin Cycle in <u>stroma</u>; 10. <u>accumulation</u> of <u>ions</u> to generate a <u>concentration gradient</u>; 11. e.g. H⁺ in <u>intermembranal space</u> in mitochondria/within <u>thylakoid space</u> in chloroplasts establishes a proton gradient for <u>chemiosmosis</u>* and formation of ATP; 12. <u>Storage of food source</u>; 13. e.g. <u>starch</u> in plant cells are stored in <u>amyloplasts</u>; R: starch grain, make sure it is membrane bound
Localisation of proteins of a related function	14. Allows <u>functionally-related proteins</u> to be <u>grouped</u> together to enhance <u>sequential biochemical processes</u>; 15. e.g. enzymes and <u>proteins</u> are grouped into <u>photosystems II and I</u> on <u>thylakoid membrane</u> of chloroplast so that electrons from

/ L	photosystem II are shuttled to photosystem I during <u>photophosphorylation</u> ;
Increase surface area / I	16. <u>Highly folded increases surface area to hold more</u> ; 17. e.g.: <u>electron transport chains/ ATP synthase in cristae of mitochondria</u> ;
Cell-cell recognition and adhesion / R	18. <u>Glycoproteins / glycolipids</u> are involved in cell-cell <u>recognition</u> ; 19. (any 1 e.g.) eg: <u>T cell receptor</u> on membranes of <u>naïve T cells</u> + how it helps to recognize <u>peptide: MC</u> on <u>antigen presenting cells</u> ;
Signal transduction / T	20. Some transmembrane proteins serve as cell surface <u>receptors</u> to transfer <u>information</u> from <u>environment</u> into cell when specific molecules (ligands) bind to them; 21. E.g. <u>glucagon</u> (hormone) bind to <u>glucagon receptor</u> in membrane which triggers a cascade of chemical reactions leading to <u>hydrolysis of glycogen to glucose</u> ; E.g. insulin
QWC: ref to 3 different contexts	

- 4 (a) Discuss how the complementary shapes between biomolecules enable eukaryotic cells to carry out their diverse functions. [13]

Category	Description
Transport protein – molecule to be transported / T	1. A <u>transport protein</u> has a <u>binding site</u> complementary in shape to <u>specific for molecule</u> , 2. E.g. <u>pyruvate – pyruvate translocase / water - aquaporin</u>
Enzyme – substrates / E	3. <u>Substrates</u> have complementary <u>conformation*</u> and <u>charge*</u> to <u>active site</u> (conformation is a more accurate word than 'shape'); 4. <u>DNA polymerase</u> – active site can bind to <u>deoxyribonucleotides</u> , 5. <u>catalyses formation of phosphodiester bonds</u> between <u>adjacent DNA nucleotides</u> to synthesise new strand (DNA replication); 6. <u>DNA ligase</u> – active site can bind to <u>deoxyribonucleotides</u> , 7. <u>forms a phosphodiester bond between two DNA fragments/ sealing nick</u> ; 8. RNA polymerase – active site can bind to <u>ribonucleotides</u> ; 9. and <u>catalyses formation of phosphodiester bonds</u> between adjacent RNA nucleotides of newly synthesised strand; 10. <u>Amino acyl RNA synthetase</u> – active site complementary to an <u>amino acid</u> and <u>tRNA</u> molecule; 11. in order to join them to form an amino acyl tRNA molecule, to help form <u>peptide bonds</u> between <u>amino acids</u> ;
Protein	12. Proteins may have <u>DNA binding domains / sites</u> that are

– nucleic acid / PN	<u>complementary in shape and charge</u> to specific <u>DNA sequence</u>; 13. <u>RNA polymerase</u> – <u>promoter</u> sequence; 14. <u>Initiating RNA synthesis/transcription</u> at <u>correct position</u> / in frame to produce functional protein; 15. <u>Transcription factors</u> - <u>specific sequence of DNA</u> adjacent to the genes that they regulate; 16. <u>Activators</u> – specific transcription factors known as <u>activators*</u> bind to <u>enhancer nucleotide sequences</u> 17. to <u>increase frequency of transcription</u>; / <u>promoting assembly of transcription initiation complex</u>; 18. <u>Repressors</u> – specific transcription factors known as <u>repressors*</u> bind to <u>silencers*</u> nucleotide sequences 19. to <u>decrease frequency of transcription</u>/ <u>inhibiting assembly of transcription initiation complex</u>; 20. <u>Spliceosome</u> recognises <u>DNA sequence at intron-exon boundary</u>, 21. to <u>excise intron and join exons to form mature mRNA</u>; 22. <u>DNA-binding proteins</u> (e.g. transcriptional repressors, histone deacetylases and repressive chromatin remodeling complexes) recognize and bind to <u>methylated DNA</u> 23. to <u>condense chromatin</u> (A: converse) results in <u>gene silencing</u>/ no gene expression;
Protein – protein / PP	24. Proteins may have <u>protein binding domains / sites</u> that are <u>complementary in shape and charge</u> to other <u>protein molecules</u>; 25. <u>Activators</u> – specific transcription factors known as <u>activators*</u> bind to <u>general transcription factors</u> at enhancers; 26. <u>promoting assembly of transcription initiation complex</u>; 27. <u>Repressors</u> – specific transcription factors known as <u>repressors*</u> bind to <u>general transcription factors</u> at silencers; 28. <u>inhibiting assembly of transcription initiation complex</u>;
Transport Or enzyme Or enzyme complex to link	29. <u>H⁺</u> must fit into <u>ATP synthase</u> 30. so that can <u>diffuse</u> through <u>ATP synthase</u>, so that ADP can be <u>phosphorylated</u> to form ATP;
Ligand - receptor / L	31. <u>Ligand</u> has a <u>complementary shape</u> that allows it to fit precisely into (extracellular domains_of) <u>receptor</u> molecules on <u>target cells</u>; 32. in order to transfer <u>information</u> from <u>environment</u> into cell when specific molecules (ligands) bind to them / convert extracellular signal to an intracellular signal; 33. Eg. Cell surface receptors: has an <u>extracellular <u>ligand-binding site*</u></u> binding to <u>hydrophilic ligand</u>, 34. Eg. <u>Binding of ligand</u> causes <u>conformational change</u> in

	<u>intracellular domain</u> of receptor; can ref to specific receptor. 35. Eg. <u>GPCR</u>: Ligand binding to GPCR causes <u>conformational change</u> in intracellular domain of receptor allowing it to bind to and <u>activate G protein</u>; or <u>RTK</u>: <u>conformational change</u>... <u>activation of kinase region</u> to result in <u>cross-phosphorylation</u>;
Immune system / I	36. <u>Ligand</u> has a <u>complementary shape</u> that allows it to fit precisely into (extracellular domains_of) <u>receptor</u> molecules on <u>target cells</u>; 37. <u>Naïve T cell</u> has specific <u>T cell receptor</u> that fits and binds to <u>peptide:MHC complex</u> presented on antigen presenting cells; 38. Helps to trigger clonal expansion and differentiation of naïve T cells; 39. <u>T helper cells</u> has specific <u>T cell receptor</u> that fits and binds to <u>specific B cell receptor</u> on <u>naïve B cells</u>; 40. Helps to trigger clonal expansion and differentiation of naïve B cells; 41. Antibodies have unique <u>antigen binding site</u> that can bind to <u>antigens/ pathogens</u> 42. causing <u>opsonisation or neutralisation</u>;
Function point (eg: points 5, 14 can only be given if named of interacting molecules are given). AVP – any valid points are awarded depending on importance of interacting molecules and depth of description. QWC: 3 different categories + overarching principle for each category	

(b) Organisms are able to detect and respond to changes in their environments. Discuss this, with reference to specific examples in unicellular and multi-cellular organisms. [12]

1. Unicellular organisms eg bacteria: responds to changes in the nutrients available in the environment in order to optimise use of resources / produce enzymes only when necessary / WTTE;
2. Specific example: ***lac operon****: genes involved in lactose metabolism under control of a single ***promoter**** and ***operator****;
3. ***Lac I*** codes for active lac repressor which binds to operator region to prevent expression of structural genes in *lac* operon;
4. Presence of lactose results in production of allolactose which binds to lac repressor to inactivate it;
5. Allowing for structural genes to be transcribed and translated to produce β -galactosidase, permease and transacetylase
6. Allows bacteria to take up lactose from its surrounding and break down lactose into glucose and galactose;
7. Presence of glucose results in low cAMP. Hence catabolite activator protein (CAP) is not activated;
8. And is therefore unable to upregulate / increase the frequency of transcription of the structural genes;
9. Specific example 2: *trp* operon: genes involved in synthesis of tryptophan under the control of a single ***promoter**** and ***operator****; (promoter and operator can be in pts 2 or 7)
10. ***Trp R*** codes for inactive *trp* repressor which cannot bind operator, allowing for transcription of structural genes in *trp* operon;

11. Tryptophan binds to trp repressor to activate it;
12. Tryptophan synthesis is thus stopped when bacteria is able to obtain sufficient amounts of tryptophan (from its environment);
13. Multicellular organisms eg humans have mechanisms to maintain a constant internal environment despite changes in the external environment;
14. Specific example: maintenance of blood glucose concentration;
15. Intake of food results in an increase in blood glucose concentration
16. which stimulates production of *insulin** by β -cells* in the *islets of Langerhan**;
17. Insulin binds to insulin receptors on *muscle cells** and *liver cells**;
18. Causes increase uptake of glucose into cells and conversion of glucose to glycogen;
19. Lack of food results in low blood glucose concentration as glucose;
20. which stimulates production of *glucagon** by *a-cells** in *islets of Langerhan**;
21. Glucagon binds to glucagon receptors on *liver cells**; R if muscle cells included
22. Causes breakdown of glycogen into glucose and release of glucose into blood;

(Signal transduction)

23. Receptor allows transduction from extracellular to intracellular;
24. Multiple steps in pathway allowing for many pts of control / different pathways in different cells to produce a coordinated response;
25. signal amplification to produce a large response;

QWC: at least 1 ref to unicellular and 1 to multicellular;

Cap unicellular to 7, multicellular to 7

A: details on nervous system

Paper 4

1

- (a) -
- (b)
- Start time / end time - did not reset stopwatch between readings, record in e.g. 00:30, 02:15;
 - Computation of time taken, in whole numbers;
 - Table 1.1: correct trend of data; in ascending order of time taken W2, 4, 1, 5, 3;
 - Table 1.2: correct trend of data; in ascending order of time taken W2, 4, 1, 5, 3;
- (c) Prepare and complete a table in the space below to show the mean time taken to reach the end-point for each glucose solution. [4]
1. Heading (glucose solution, mean time taken with correct units (s not sec))
 2. Correct Calculation of mean;
 3. Mean time taken recorded to nearest second / 1 d.p.;
 4. Table must show all grid lines and border, drawn using ruler.
- (d) A sixth boiling tube should be set-up. Describe how a control is set-up and explain the reason for this. [2]
1. With all other variables kept constant, replace glucose solution is with distilled water;
 2. This is to show that glucose is responsible for reduction and hence the decolourisation / colour change of permanganate;
- (e) Present the above data in a graphical format. [5]
1. S – Scale must be appropriate and both axis must cover at least 1/2 of the space given.
 2. A – correct choice of axes + correct labels and units + correctly labelled intervals
 3. P - all points plotted correctly
 4. L - all points joined by line of best fit and
 5. E - no extrapolation beyond plotted points
- (f) Estimate the concentration of,
- (i) solution **W1** by using the graph
- Concentration of solution **W1** :[1]
- Read off graph + show dotted line on graph and values on x and y axes
+ Value should be within teachers' range.
- Without extrapolating the graph, estimate the concentration of solution **W5** by :
- (ii) Using the formula $y=mx + c$,. Show your working clearly, on how the values of m and c are calculated.
- Concentration of solution U :[3]
- 1) Use values from graph to calculate gradient ;

$$m : \text{Gradient} = \frac{115 - 10}{5 - 25}$$

$$= - 5.25$$
 - 2) Use gradient (m) and y value from graph to calculate C ; Reject if candidate use $y = 0$

$$C : Y = -5.25x + C$$

$$88 = -5.25(10) + C$$

$$C = 140.5$$
 - 3) Use calculate gradient (m), value of C and time taken for wW to turn from pink to colourless to calculate concentration. Values of time take for W5 to turn colourless must be within teachers' value.

$$X : \text{substituting } y = mx + C ,$$

$$77.5 = -5.25x + 140.5$$

$$X = 12 \%$$

2 You are given a prepared slide showing a section of a root tip showing different stages of mitosis.

(a) Examine the slide. Identify 2 cells at different stages of mitosis and make a large labelled drawing of each cell. Indicate the stage of mitosis for each of the cells drawn. [6]

1. Drawing of chromosomes matched with label of stage and accurately represented (cell 1)
2. Drawing of chromosomes matched with label of stage and accurately represented (cell 2)
 - * Reject if individual chromosomes are drawn or if drawing is inaccurate. E.g. nuclear enveloped has not disintegrated in Metaphase or Anaphase.
 - * Reject interphase. Not part of mitosis.
3. Sharp and smooth continuous line. Reject fuzzy lines, or messy drawing with blunt pencil
4. Proportion: Thin cell wall of consistent thickness + size of chromosomes
5. Labels: cellulose cell wall, plasma membrane, chromosome, cytoplasm (any 3 labelled)
6. Large: 2/3 of space provided

(b) Calculate the actual length of the chromosome shown in Fig. 2.1 in μm .

..... μm [3]

10 eyepiece graticule division = 0.02mm = 20 μm ; (conversion of mm to μm)

1 eyepiece graticule division = 20/10 = 2 μm ; (calibrate eyepiece graticule)

A: alternative definitions of a "division" = smallest division as 1 according to diagram or smallest division = $\frac{1}{4}$

1 division = 20/40 = 0.5 μm

Length of chromosome = 1.5 eyepiece graticule = 3 μm ; (measurement of chromosome)

(c) Using the prepared slide, count the number of cells in each stage of the mitotic cell cycle, using 100 cells from a randomly chosen field of view of the microscope.

Repeat the count for another chosen field of view.

(i) Prepare and complete a table in the space below to show the percentage of cells in each stage of the mitotic cell cycle. [5]

Stage	Number of cells out of 100			Percentage of cells in stage / %
	Count 1	Count 2	Average	
Interphase				
Prophase				
Metaphase				
Anaphase				
Telophase				

1. Appropriate headings with units
2. All stages listed
3. Cells out of 100 + average – recorded to nearest whole number
4. Percentage of cells to whole number
A and %: up to 1 d.p.
5. Correct trend: Interphase most, Meta/Ana - least

(b) The percentage of cells observed in each stage can be used to infer the duration of the stage.

(i) State the longest stage of the cell cycle and explain how you arrived at your conclusion. [2]

1. Interphase; (ecf – based on student's results)
2. Majority / Higher percentage of cells in stage implies that the stage took the longest duration to complete hence highest probability of finding a cell in that stage;

(ii) A student commented that a vital piece of information is required to determine the exact duration of each stage. Suggest what is this vital piece of information. [1]

- Exact timing for completion of cell cycle
- (iii) Another student commented that this method of determining the duration of each stage is inaccurate.

Other than the absence of 3 replicates, suggest why it is so and describe how you can improve on the accuracy of the results obtained. [2]

Error:

- Small numbers (100) of cells counted / small number cells in metaphase / anaphase / telophase / not all cells in each stage may be observed, hence not accurate;

How to improve:

- Count larger number of cells eg. 200 (must specify at least 200) / different root samples or slides ;

- (c) Using the formula above and the probability table given, what conclusion can be drawn from this t-test? [3]

1. Degree of freedom = $10 + 10 - 2 = 18$, from table: $p < 0.001$;
2. Since $p < 0.05$, at 5% level of significance,
3. the difference between T1 and T2 is significant and not due to chance;
4. Rate of cell division is higher at higher temperature;

- 3 It is known that Q, an organic compound can act as an inhibitor of phosphatase. Using the list of apparatus below, design an experiment to determine if Q is a competitive or a non-competitive inhibitor. [14]

Aim

To determine if Q is a competitive or a non-competitive inhibitor of phosphatase activity.

Theory

- T1. Substrate, phenolphthalein diphosphate (XP) complementary in shape and charge (must have both shape and charge) to **active site*** of phosphatase;
- T2. Intensity of colour/pink compound can be measured using a spectrophotometer over a fixed period of time is an indication of phosphatase activity;
- T3. If Q is a competitive inhibitor, it competes with substrate for **active site***;
- T4. If Q is a non-competitive inhibitor, it binds to a site other than active site, causing a change in active site;

[Theory T1, T2, max 2 marks]

Trend (Expected results / how to make conclusion)

- Tr1. If Q is a competitive inhibitor, same maximum absorbance reached at high [XP], compared to without inhibitor;
- Tr2. If Q is a non-competitive inhibitor, lower maximum absorbance at high [XP], compared to without inhibitor

[Trend Tr, max 1 mark]

Variables

	Description	How it is controlled
Independent variable	IV1: <u>Presence and absence of inhibitor</u>	• Conduct experiment, <u>with and without inhibitor Q</u> .
	IV2: <u>5 concentrations of XP</u>	• Dilution table (5 regular [XP]).
Measurable variable	MV: <u>Measure absorbance</u>	• <u>Measure 1 cm³ of solution (XP + phosphatase + sodium carbonate), add it into a cuvette.</u>
Dependent variable	DV: <u>Rate of phosphatase activity</u>	• obtained by measuring <u>absorbance of pink colouration divided by time</u> (idea of fixed time).

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[Independent variable – IV1: with and without inhibitor Q, IV2: 5 concentrations of XP in dilution table]

[Measurable variable – state + how it is measured (MV)]

[Dependent variable – state + how it is measured (DV)]

Conduct a pilot experiment to determine suitability of apparatus, range of independent variables (e.g. range of concentration of inhibitor Q), duration of experiment & amount of materials used.

Numbered steps

Preparation of inhibitor solution

- Using 10 cm³ syringe, prepare 5 different concentrations of XP solution by following dilution table below:

Concentration of XP /%	Volume of 10% XP used /cm ³	Volume of distilled water used /cm ³
2	2.0	8.0
4	4.0	6.0
6	6.0	4.0
8	8.0	2.0
10	10.0	0.0

Preparation of enzyme reaction

- Set up a water bath of 37°C, using hot and cold water, and use a thermometer to monitor temperature.
(constant variable: temperature
+ how: using hot and cold water, use a thermometer to monitor temperature
+ why: temp. affect rate of enzyme catalyzed reactions)
- Place substrate XP, enzyme phosphatase and inhibitor Q, separately into water bath. Allow 1 min of equilibration time, so that solutions can attain temperature of water bath.
[Equilibration E]
- Using a syringe add 1 cm³ of 2%, 4%, 6%, 8% and 10% XP to test tubes.
- Add 1 cm³ of 5% inhibitor Q to each test tube.
- Add 1 cm³ of pH buffer 7 to each test tube.
(constant variable: pH
+ how: using buffer/ fix pH eg pH7
+ why: pH changes denatures enzymes)
- Add 1 cm³ of 5% phosphatase to each test tube. Mix well and incubate mixture in water bath.
(constant variable: concentration of enzyme / volume of enzyme
+ how: using syringe / fix vol. eg 1 cm³
+ why: affect total amount of enzyme / final [enzyme] , so affects rate of reaction)
- Prepare a “blank” – fill a cuvette with 1 cm³ distilled water and measure its absorbance value using a spectrophotometer. This acts a blank / absorbance should be at 0% / if it is not 0% then spectrophotometer should be re-zeroed.
[Blank – state how + why, B]
- After 5 minutes, add 1 cm³ of mixture with 1 cm³ sodium carbonate to get a pink colouration.
(constant variable: reaction time)
- Transfer 1 cm³ of reaction mixture into a cuvette and measure absorbance value/colour intensity of pink solution using a spectrophotometer.

11. Repeat Step 3-10 using distilled water instead of inhibitor Q.
12. Record colour intensity in absorbance value (A) for each concentration of XP.
13. Plot a graph of colour intensity in absorbance value/A against concentration of XP.

[Constant variable: state any 2]

[Constant variable: how + why it is controlled CV1]

[Constant variable: how + why it is controlled CV2]

Control

- Keep all variables constant, set up control experiment using boiled and cooled phosphatase/ distilled water instead of inhibitor Q.
- This is to obtain baseline results, in absence of inhibitors/ to show phosphatase enzyme is responsible for dephosphorylation of XP.

[Control: state + purpose, Co]

Repeats & replicates

- Conduct 3 replicates for each condition to check that no anomalies are present.
- Repeat entire experiment twice to check for reproducibility.

[Repeat and replicates: state how + why purpose, RR]

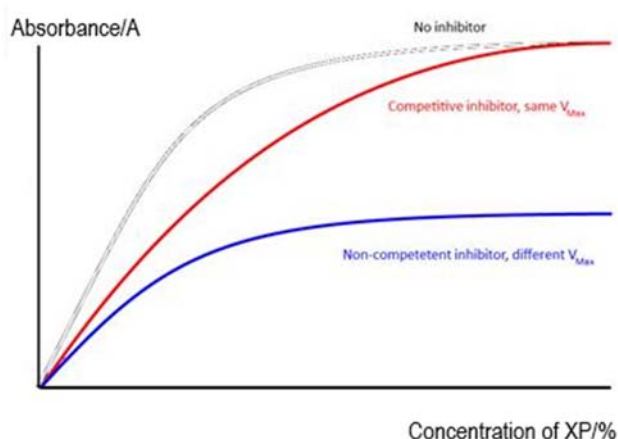
Data Recording and Processing

Record and process data: (table and graph should have same heading / unit)

Concentration of XP /%	Absorbance /A							
	With inhibitor Q				Without inhibitor Q			
	Trial 1	Trial 2	Trial 3	Average	Trial 1	Trial 2	Trial 3	Average
2								
4								
6								
8								
10								

[Table, Ta]

Graph on effect of inhibitor Q on rate of phosphatase activity at various concentration of substrate



[Graph, G: shows expected trend]

Risks & precautions

1. Wear gloves and goggles when handling XP solution, inhibitor Q solution and phosphatase solutions as they may be toxic.

2. Do not touch electrical switch of the spectrophotometer with wet hands to prevent electrocution.
 3. Wear oven mittens / insulating gloves when handling hot water to avoid scalding.
- [Risk and Precautions, RP: state 1 risk + precaution]**

Compulsory point to get max 14m: Answering the question